



C5

The physical world, and choosing

30 minutes

BY THE END OF THIS CAPSULE YOU WILL BE ABLE TO

Pick the right kind of AI for a problem you have never seen before — and justify the choice.



The last one: machines that move

Physical or embodied AI — and the widest gap between the demo and the world.

89.4%

in simulation

RLBench — rigid objects, known positions, perfect information

12%

in a real house

Real household tasks. Same field. Different universe.

Why the real world is different

- No undo — A wrong token is regenerated. A dropped plate is not.
- Infinite long tail — Every house is different. No two piles of laundry are alike.
- Data cannot be scraped — Every demonstration is collected in real time, by a person.

But where the environment is constrained, it already works: Waymo ~450,000 trips a week; Apollo Go 11 million driverless rides in China, up 175%.

Source: Stanford HAI, AI Index Report 2026



Six kinds of AI, and how each one fails

Type	What it answers	What it needs	How it fails
Discriminative ML	Which bucket is this?	Thousands of labels	Anything unfamiliar
Generative AI	What comes next?	Scale and compute	Confidently, and often
Foundation models	Almost anything	Enormous scale	Counting, exactness
Agents	Can you go and do it?	Tools and a loop	1 in 3 attempts
Multimodal	What do you see or hear?	Paired data	Fine detail
Physical AI	Can you move it?	The real world	88% of home tasks

Photograph this slide. It is the whole of Class 1 on one page.



Four questions, in this order

Ask them of any problem, before you pick anything.

1

What kind of output do you need?

A label · new content · an action · a physical act

Two of these decide it outright. The other two hand you to Q2.

2

How much labelled data do you have?

Thousands → a classifier is viable. None → foundation model.

This is a question about your organisation, not the technology.

3

What does being wrong cost — and who does it hit?

One person, a team, or everyone. Trivial, or irreversible.

The two axes Moderna arrived at independently.

4

Can you check the answer?

Verifiable → you can trust it more. Unverifiable → you cannot.

If you cannot check it, you cannot deploy it unsupervised.



POLL · FIVE YOU HAVE NOT SEEN

Which of these can today's AI do reliably?

New tasks. Use the four questions.

Summarise a 100-page contract

Tell you exactly how many pages it has

Spot fraud across 10 million transactions

Assemble flat-pack furniture

Diagnose a rare disease from symptoms

Three hours ago you guessed. This time you have a method.

Wooclap results



And here is why

Each one is decided by a different question from the framework.

Summarise a 100-page contract	ACES	Q1 — the output is language, and there is abundant data
Tell you exactly how many pages	FAILS	Q1 — that is an exact discrete count, not language
Spot fraud in 10M transactions	ACES	Q2 — you have millions of labels. Use a classifier, not a model
Assemble flat-pack furniture	FAILS	Q1 — a physical act, and demonstrations cannot be scraped
Diagnose a rare disease	DEPENDS	Q3 — rare means thin data, and being wrong is irreversible

You could not have reasoned your way to those three hours ago. Now you can.



What Moderna actually did

They asked question 3 — and turned it into a grid.

AUDIENCE	IMPACT OF FAILURE		
	Low	Medium	Critical
	Individual		
	Team		
Company		Benefits Assistant	DoseID

- Minimal controls
- Quality, maintenance, support
- Cyber review · quarterly oversight · documented design

Source: HBS 625-070, Exhibit 11

Not “centralise or don’t”. Sort the use cases by what being wrong costs.

POLL · THE SAME QUESTION AGAIN

Should Moderna take centralised control of employee-built GPTs?

A Yes — centralise all of them

B No — keep it open

C Only the high-risk ones

D Not sure

Wooclap results

Identical wording to the vote you took forty minutes ago.



What changed in forty minutes

Same question. Same room. Nothing about Moderna changed.

BEFORE THE FRAMEWORK

[paste the C4 result here]

AFTER

[paste the C5 result here]

Nothing about the company changed in forty minutes. You did. That is what a framework is for.



Coming in two weeks: A1

16% · individual

Everything you did today, turned into work: stop arguing about which approach is better — build both, and measure.

1 Choose the right tool

One classification task, built twice — a trained classifier, and a prompt. Measure accuracy, cost, latency, effort. Which would you deploy, and at what scale does that change?

2 Build a retrieval system

10–30 documents from a domain you know. Answer five questions it could not answer without them — then break it, and explain why it broke.

3 Prompt and evaluate

Write a prompt. Build ten test cases. Run it, change the prompt once, run it again. Report what moved.

Half the marks are for thinking, not coding. Technical execution is 30%. Understanding and reasoning together are 50%. Starter notebooks provided for all three parts.

Released Sun 16 Aug

Due Sun 30 Aug, 23:59

Full brief on NTULearn

Pick a domain you actually know — your degree subject, a hobby, a job, a city, a sport. It does not have to be business.



Three hours ago

you guessed. Now you have a method.

1 The frontier is jagged

Capability does not track how hard a task looks. You have to test.

2 Six kinds of AI

Each answers a different question, and each fails differently.

3 Four questions

Output · data · cost of being wrong · can you check it.

4 One live decision

Moderna's — and you answered it better at the end than at the start.

NEXT WEEK The modern AI stack — how any of this actually becomes a product.

BEFORE THEN Week 2 readings on NTULearn — lighter than this week. A1 releases Sun 16 Aug.



Appendix — where the framework breaks

Not part of the 30 minutes. The honest limits of what you just learned.

It assumes you know the cost of being wrong

Often you do not, until it has gone wrong. Moderna did not classify the Self-Review GPT as high-risk until its usage spiked and someone asked what it was displacing.

It says nothing about whether you should

Every question here is about capability and risk. None asks whether the task should be automated at all — Franklin's objection to the Self-Review GPT was developmental, not technical, and this framework would have missed it entirely. Class 6 takes that up.

The boundaries move

Question 2 assumed you need thousands of labels for a classifier. Few-shot methods erode that. Question 1 assumed exact counting is impossible — tool use fixes it. Re-run the framework annually; the answers change even when the questions do not.

